KGiSL INSTITUTE OF INFORMATION MANAGEMENT

Approved by *AICTE*, New Delhi and Affiliated to Bharathiar University

ISO 9001:2015 Certified Institution

KGiSL Campus, Saravanampatti, Coimbatore – 641 035

KGiSL-IIM

co-Kreate your Genius

Master of Computer Applications

Year 20 - 20

Department of Computer Science

KGiSL INSTITUTE OF INFORMATION MANAGEMENT

COIMBATORE-641 035

KGiSL INSTITUTE OF INFORMATION MANAGEMENT
COIMBATORE - 641 035
(Affiliated to Bharathiar University)

DEPARTMENT OF COMPUTER SCIENCE

LABORATORY RECORD

NAME: _____ **ROLL No.:** _____

SEMESTER: _____ **YEAR:** _____

SUBJECT: _____

Certified Bonafide Record of Work Done by

Mr / Ms _____

Staff In Charge HOD

Place : _____

Date : _____

University Register No. _____

Submitted for the University Practical Examination held on _____

Internal Examiner External Examiner

INDEX

S.No	Date	Title	Page No.	Signature
1		Image Slideshow using JavaScript		
2		Electricity Bill Calculation using PHP		
3		Web-Based Feedback System using PHP and MySQL		
4		User Login using PHP and MySQL		
5		File Handling using Node.js		
6		Student Registration Form using Node.js and Express.js		
7		Employee Information Portal using Node.js, Express.js and MongoDB		
8		Git Repository: Basic Version Control Operations		

1. Image Slideshow using JavaScript

Source Code

Index.html

```
        <!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Image Slideshow</title>
    <link rel="stylesheet" href="./style.css" />
</head>
<body>
   <img src=".//assets/img1.jpg" alt="no iamge"    id ="slide_show"    class="image"/>
<script src="./app.js"></script>
</body>
</html>
```

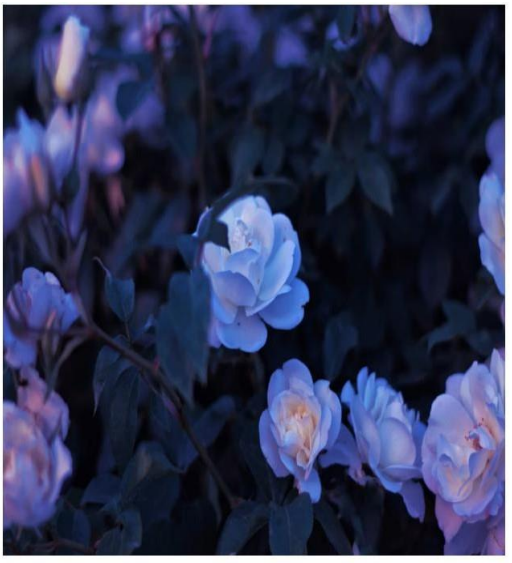
CSS

```
.image{
    height:500px;
    width:500px;
    margin:auto;
    display: flex;
    justify-content:center;
}
```

App.js

```
let images=[
    "./assets/img1.jpg",
    "./assets/img2.jpg",
    "./assets/img3.jpg",
    "./assets/img4.avif"
];
let index=0;
let slide = document.getElementById('slide_show');
let changeimg=()=>{
    index++;//index=index+1
    if(index >=images.length){
        index=0;
    }
    slide.src =images[index];
}
setInterval(changeimg,2000);
```

output



2. Electricity Bill Calculation using PHP

Source code

```
<?php
// Initialize variables
$show_bill = false;
$name = "";
$units = "";
$energy_charges = 0;
$fixed_charges = 50.00;
$total_amount = 0;

if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $name = htmlspecialchars($_POST['consumer_name']);
    $units = (int)$_POST['units'];

    // Calculation Logic
    // First 100 units @ ₹ 2.50
    // Remaining units @ ₹ 3.50
    if ($units <= 100) {
        $energy_charges = $units * 2.50;
    } else {
        $energy_charges = (100 * 2.50) + (($units - 100) * 3.50);}

    $total_amount = $energy_charges + $fixed_charges;
    $show_bill = true;
}
?>

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Electricity Bill Generator</title>
    <style>
        body {
            background-color: #eaf5fa;
            font-family: Arial, sans-serif;
            color: #00334e;
            display: flex;
            flex-direction: column;
            align-items: center;
            padding-top: 40px;
            margin: 0;
```

```
}

h1 {
    color: #003b5c;
    margin-bottom: 30px;
}
.form-table {
    border-spacing: 0 15px;
}
.form-table td {
    padding: 0 10px;
}
.form-table label {
    font-weight: bold;
    color: #003b5c;
}
.form-table input {
    padding: 8px;
    border: 1px solid #a8c6d9;
    border-radius: 3px;
    width: 170px;
    font-size: 14px;
}
.btn-container {
    text-align: center;
    margin-top: 10px;
}
.submit-btn {
    padding: 6px 15px;
    background-color: #f7fbfc;
    border: 1px solid #7094a8;
    color: #003b5c;
    border-radius: 3px;
    cursor: pointer;
    font-size: 14px;
}
.submit-btn:hover {
    background-color: #dbeaf1;
}

.bill-container {
    margin-top: 40px;
    background-color: #ffffff;
```

```
    border: 2px solid #1c4b66;
    padding: 30px 40px;
    width: 420px;
    box-sizing: border-box;
    box-shadow: 2px 2px 5px rgba(0,0,0,0.05);
}
.bill-title {
    text-align: center;
    font-size: 20px;
    font-weight: bold;
    color: #003b5c;
    text-transform: uppercase;
    margin-bottom: 20px;
}

hr {
    border: 0;
    border-top: 1px solid #cbd6dd;
    margin: 15px 0;
}

.bill-details {
    width: 100%;
    border-collapse: collapse;
}
.bill-details td {
    padding: 8px 0;
    font-size: 15px;
}
.bill-details td:first-child {
    font-weight: bold;
    color: #003b5c;
    width: 60%;
}
.bill-details td:last-child {
    color: #2c5973;
}

.total-row {
    display: flex;
    justify-content: space-between;
    padding: 5px 0;
    color: #922036;
    font-size: 17px;
```

```
            font-weight: bold;
        }

        .footer-msg {
            text-align: center;
            font-size: 13px;
            font-weight: bold;
            color: #003b5c;
            margin-top: 20px;
        }
    </style>
</head>
<body>

    <h1>Electricity Bill Generator</h1>

    <form method="POST" action="">
        <table class="form-table">
            <tr>
                <td align="right"><label for="consumer_name">Consumer Name</label></td>
                <td><input type="text" id="consumer_name" name="consumer_name"
value="<?php echo htmlspecialchars($name); ?>" required></td>
            </tr>
            <tr>
                <td align="right"><label for="units">Units Consumed</label></td>
                <td><input type="number" id="units" name="units" value="<?php echo
htmlspecialchars($units); ?>" required min="0"></td>
            </tr>
        </table>
        <div class="btn-container">
            <button type="submit" class="submit-btn">Generate Bill</button>
        </div>
    </form>
    <?php if ($show_bill): ?>
    <div class="bill-container">
        <div class="bill-title">TNEB ELECTRICITY BILL</div>
        <hr>
        <table class="bill-details">
            <tr>
                <td>Consumer Name</td>
                <td><?php echo $name; ?></td>
            </tr>
```

```
        <tr>
            <td>Units Consumed</td>
            <td><?php echo $units; ?> Units</td>
        </tr>
        <tr>
            <td>Energy Charges</td>
            <td>₹ <?php echo number_format($energy_charges, 2); ?></td>
        </tr>
        <tr>
            <td>Fixed Charges</td>
            <td>₹ <?php echo number_format($fixed_charges, 2); ?></td>
        </tr>
    </table>

    <hr>
    <div class="total-row">
        <span>Total Bill Amount</span>
        <span>₹ <?php echo number_format($total_amount, 2); ?></span>
    </div>
    <hr>
    <div class="footer-msg">Thank You - Pay Before Due Date</div>
  </div>
  <?php endif; ?>
</body>
</html>
```

Output


Electricity Bill Generator
Consumer Name
batman
Units Consumed
1000
Generate Bill
TNEB ELECTRICITY BILL
Consumer Name
batman
Units Consumed
1000 Units
Energy Charges
₹ 3,400.00
Fixed Charges
₹ 50.00
Total Bill Amount
₹ 3,450.00
Thank You - Pay Before Due Date

3. Web-Based Feedback System using PHP and MySQL

Source Code

Db.php

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
   <?php
   $servername ="localhost";
   $username="root";
   $password="";
   $database = "feedbackdb";
   $conn=mysqli_connect(
    $servername,
    $username,
    $password,
    $database
   );

   if($conn){
    echo "Connection  Successfully";

   }else{
    echo "Connection failed";
   }

    ?>
</body>
</html>
```

Index.php

```
<?php
include("db.php");

if(isset($_POST['submit'])){

    $name = $_POST['name'];
    $email = $_POST['email'];
    $message = $_POST['message'];
```

```
    $sql = "INSERT INTO feedback(name, email, message)
            VALUES('$name', '$email', '$message')";

    if(mysqli_query($conn, $sql)){
        echo "<script>alert('Feedback submitted Successfully');</script>";
    }
    else{
        echo "Error: " . mysqli_error($conn);
    }
}
?>

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Feedback System</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f2f2f2;
            margin: 0;
            padding: 0;
        }

        form {
            width: 400px;
            margin: 80px auto 20px;
            padding: 30px;
            background-color: white;
            border-radius: 10px;
            box-shadow: 0 0 10px rgba(0,0,0,0.2);
        }

        h2 {
            text-align: center;
            margin-bottom: 25px;
        }

        label {
            display: block;
```

```
    margin-top: 15px;
    margin-bottom: 6px;
    font-weight: bold;
}

input[type="text"],
textarea {
    width: 100%;
    padding: 10px;
    border: 1px solid #ccc;
    border-radius: 5px;
    box-sizing: border-box;
    font-size: 14px;
}

textarea {
    resize: vertical;
}

input[type="submit"] {
    width: 100%;
    padding: 12px;
    margin-top: 20px;
    border: none;
    border-radius: 5px;
    background-color: #333;
    color: white;
    font-size: 16px;
    cursor: pointer;
}

input[type="submit"]:hover {
    background-color: #555;
}

a{
    display: block;
    width: 400px;
    margin: 10px auto;
    text-align: center;
    text-decoration: none;
    color: #333;
    font-weight: bold;
}
```

```
        a:hover {
            text-decoration: underline;
        }
    </style>
</head>
<body>

    <form action="" method="post">

        <h2>Feedback Form</h2>
        <label>Name</label>
        <input type="text" name="name" required>

        <label>Email</label>
        <input type="text" name="email" required>
        <label>Feedback</label>
        <textarea name="message" rows="5" required></textarea>

        <input type="submit" name="submit" value="Submit Feedback"/>
    </form>
    <a href="./view.php">View Data</a>

</body>
</html>
```

View.php

```
<?php
include("db.php");
?>
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>View Feedback</title>

    <style>
        * {
```

```
    box-sizing: border-box;
    margin: 0;
    padding: 0;
    font-family: Arial, sans-serif;
}
body {
    background: #f4f6f9;
    padding: 40px 20px;
}

.container {
    max-width: 1000px;
    margin: auto;
    background: white;
    padding: 30px;
    border-radius: 12px;
    box-shadow: 0 5px 20px rgba(0,0,0,0.1);
}
h1 {
    text-align: center;
    color: #333;
    margin-bottom: 25px;
}
table {
    width: 100%;
    border-collapse: collapse;
    margin-bottom: 25px;
}
th {
    background: #333;
    color: white;
    padding: 14px;
    text-align: left;
}

td {
    padding: 12px 14px;
    border-bottom: 1px solid #ddd;
    color: #444;
}
```

```
        tr:hover {
            background: #f5f5f5;
        }

        .back-btn {
            display: inline-block;
            text-decoration: none;
            background: #333;
            color: white;
            padding: 10px 18px;
            border-radius: 6px;
        }

        .back-btn:hover {
            background: #555;
        }

        @media (max-width: 700px) {
            .container {
                padding: 15px;
                overflow-x: auto;
            }

            table {
                min-width: 600px;
            }
        }
    </style>
</head>

<body>
<div class="container">
    <h1>Feedback Details</h1>

    <table>
        <tr>
            <th>ID</th>
            <th>Name</th>
            <th>Email</th>
            <th>Message</th>
        </tr>
```

```
        <?php

        $sql = "SELECT * FROM feedback";
        $result = mysqli_query($conn, $sql);

        while($row = mysqli_fetch_assoc($result)) {

        ?>

        <tr>
            <td><?php echo $row['id']; ?></td>
            <td><?php echo $row['name']; ?></td>
            <td><?php echo $row['email']; ?></td>
            <td><?php echo $row['message']; ?></td>
        </tr>

        <?php
        }
        ?>

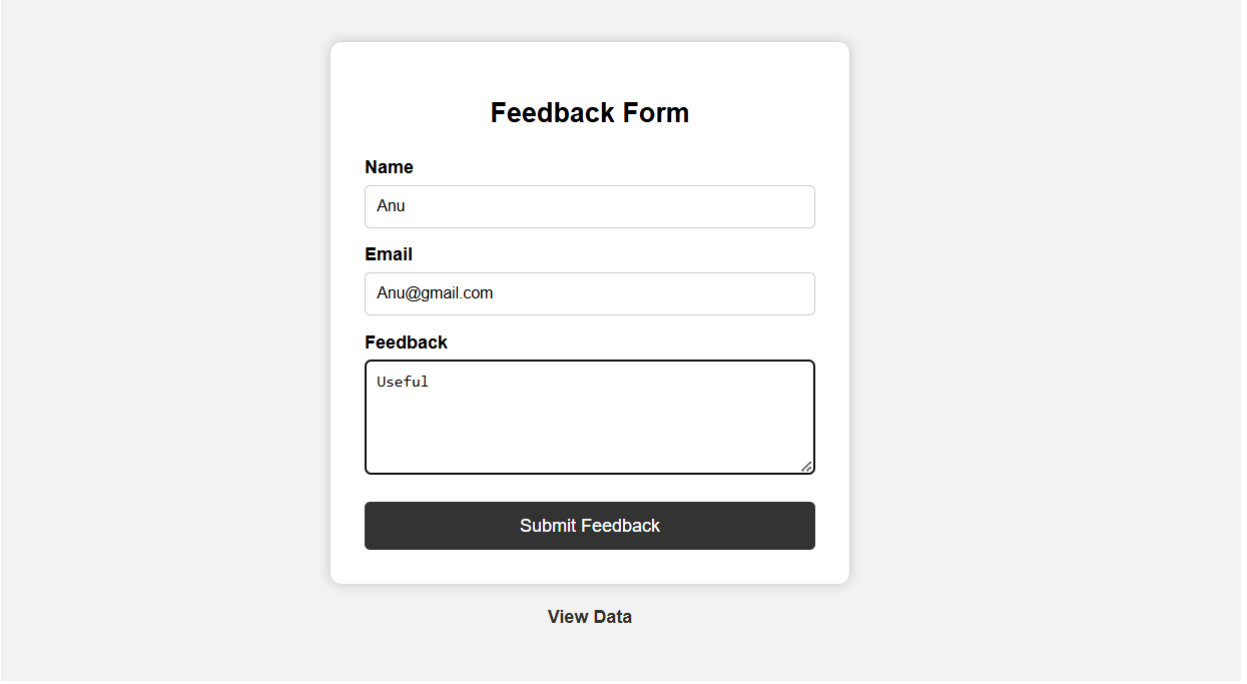
    </table>

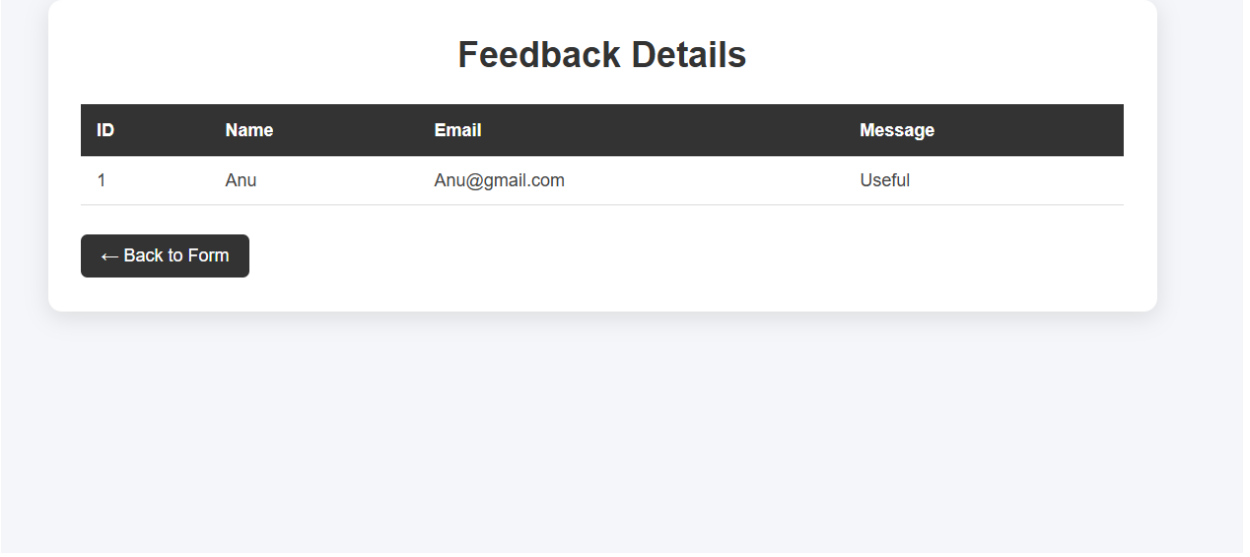
    <a href="./index.php" class="back-btn">← Back to Form</a>

</div>

</body>
</html>
```

Output


Feedback Form
Name
Anu
Email
Anu@gmail.com
Feedback
Useful
Submit Feedback
View Data


Feedback Details
ID
Name
Email
Message
1
Anu
Anu@gmail.com
Useful
← Back to Form

4. User Login Verification using PHP and MySQL

Source Code

Login.php

```
<?php

session_start();
include('db.php');

if(isset($_POST['submit'])) {
   $username = $_POST['username'];
   $password = $_POST['password'];

   $sql = "SELECT * FROM user
         WHERE username='$username'
         AND password='$password'";
   $result = mysqli_query($conn, $sql);

   if(mysqli_num_rows($result) == 1) {

      $_SESSION["username"] = $username;
      header("Location: dashboard.php");
      exit();

   } else {
      $error = "Invalid username or password";
   }
}
?>

<!DOCTYPE html>
<html lang="en">

<head>

<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>User Login</title>

<style>

* {
    box-sizing: border-box;
    font-family: Arial, sans-serif;
}
body {
    margin: 0;
    background: #f5f3e8;
    min-height: 100vh;

    display: flex;
    justify-content: center;
    align-items: center;
}

.login-box {
    width: 380px;
    background: white;
    padding: 35px;
    border-radius: 10px;
    box-shadow: 0 5px 20px rgba(0,0,0,0.12);
}
h2 {
    text-align: center;
    color: #315c45;
    margin-bottom: 25px;
}

label {
    display: block;
    margin-bottom: 7px;
    color: #444;
}

input[type="text"],
```

```
input[type="password"] {
    width: 100%;
    padding: 11px;
    margin-bottom: 18px;
    border: 1px solid #ccc;
    border-radius: 5px;
}

input:focus {
    outline: none;
    border-color: #315c45;
}

.login-btn {
    width: 100%;
    padding: 12px;
    background: #315c45;
    color: white;

    border: none;
    border-radius: 5px;
    cursor: pointer;
    font-size: 15px;
}

.login-btn:hover {
    background: #264936;
}

.error {
    background: #f8dddd;
    color: #a33a3a;
    padding: 10px;
    margin-bottom: 18px;

    border-radius: 5px;
    text-align: center;
}
```

```
</style>

</head>
<body>

<div class="login-box">

    <h2>User Login</h2>
    <?php if(isset($error)) { ?>

        <div class="error">
            <?php echo $error; ?>
        </div>
    <?php } ?>

    <form method="post">

        <label>Username</label>
        <input
            type="text"
            name="username"
            placeholder="Enter username"
            required>
        <label>Password</label>
        <input
            type="password"
            name="password"
            placeholder="Enter password"
            required>

        <input
            type="submit"
            name="submit"
```

```
        value="Login"
        class="loginbtn>
    </form>

</div>

</body>
</html>
```

Logout.php

```
<?php

session_start();

session_destroy();

header("Location: login.php");

exit();

?>
```

Dashboard.php

```
<?php

session_start();

if(!isset($_SESSION['username']))  {

    header("Location: login.php");
    exit();
}
?>

<!DOCTYPE html>
<html lang="en">

<head>
```

```
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Dashboard</title>

<style>

* {
    box-sizing: border-box;
    font-family: Arial, sans-serif;
}

body {
    margin: 0;
    background: #f5f3e8;
    min-height: 100vh;
    display: flex;
    justify-content: center;
    align-items: center;
}
.dashboard {
    width: 420px;
    background: white;

    padding: 40px;
    text-align: center;
    border-radius: 10px;

    box-shadow: 0 5px 20px rgba(0,0,0,0.12);
}

.dashboard h1 {
    color: #315c45;
    margin-bottom: 10px;
}

.dashboard p {
    color: #666;
```

```
    margin-bottom: 30px;
}

.logout {
    display: inline-block;

    padding: 11px 30px;

    background: #315c45;
    color: white;
    text-decoration: none;

    border-radius: 5px;
}

.logout:hover {
    background: #264936;
}

</style>

</head>
<body>

<div class="dashboard">

    <h1>
        Welcome, <?php echo $_SESSION['username']; ?>
    </h1>

    <p>
        You have successfully logged in.
    </p>
    <a href="logout.php" class="logout">
        Logout
    </a>

</div>
```

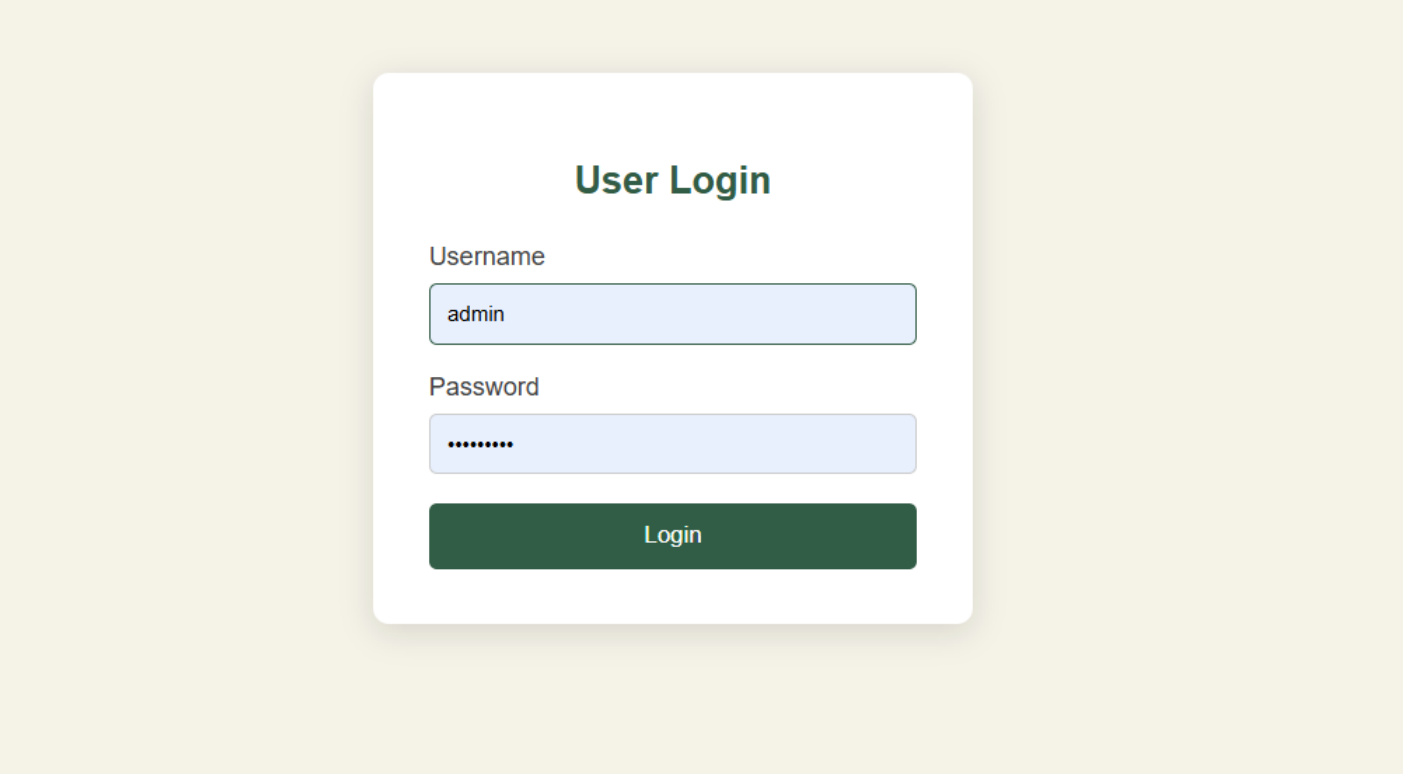
```
</body>

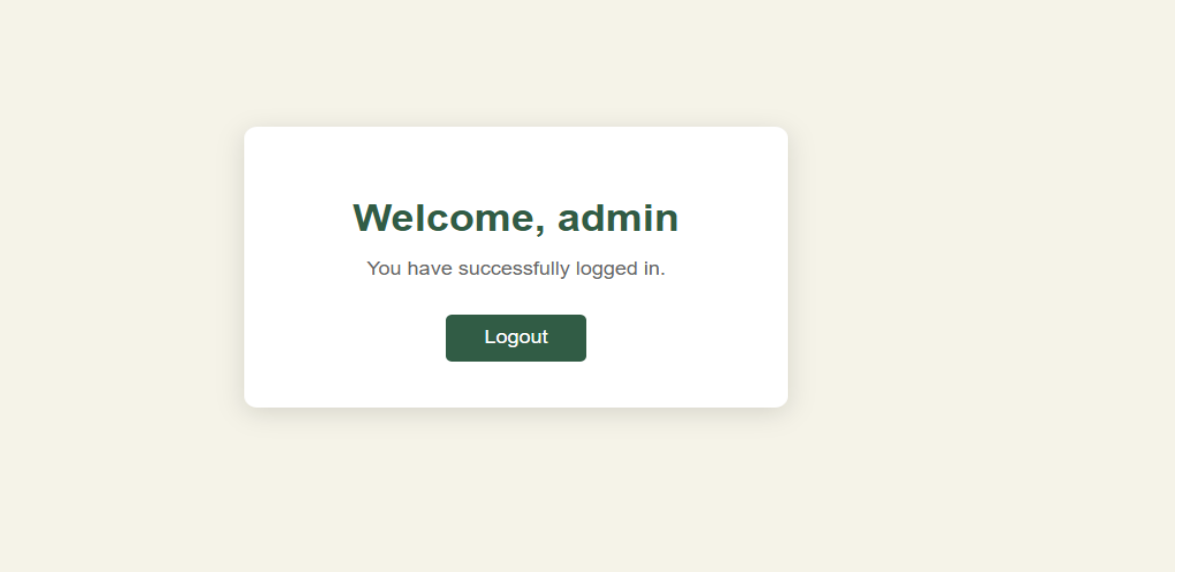
</html>
```
Db.php
```
<?php

$servername = "localhost";
$username = "root";
$password = "";
$database = "login";
$port = 3308;

$conn = mysqli_connect(
    $servername,
    $username,
    $password,
    $database,
    $port
);
if (!$conn) {
    die("Connection failed");
}
?>
```

Output


User Login
Username
admin
Password
Login


Welcome, admin
You have successfully logged in.
Logout

5. File Handling using Node.js

Source code

```
const fs=require('fs');
const file1 ="page.text";
const data=`
Employee Name: 'pavithra',
Employee ID: '001';
fs.writeFile(file1,data,function(
err)
{
   if(err){
      console.log(err);
   }else{
      console.log("File created");
   }
   fs.readFile(file1,(err,data)=>{
      if(err){
         console.log(err);
      }else{
         console.log(data);
      }
   })
   console.log(data);
});
```

Output


PS C:\Xampp\htdocs\program5> node app.js
File created
Employee Name: 'pavithra',
Employee ID: '001'

6. Student Registration Form using Node.js and Express.js

Source code

Login.html

```
<!doctype html>
<html>
  <head>
    <title>Student Registration - Login</title>
    <link rel="stylesheet" href="/style.css" />
  </head>

  <body>
    <div class="container">
      <h1>Student Registration System</h1>
      <h2>Login</h2>

      <form action="/login" method="POST">
        <label>Username</label>
        <input type="text" name="username" required />
        <label>Password</label>
        <input type="password" name="password" required />
        <button type="submit">Login</button>
      </form>
      <p class="info">
        Username: <b>admin</b><br />
        Password: <b>1234</b>
      </p>
    </div>
  </body>
</html>
```

Register.html

```
<!doctype html>
<html>
  <head>
    <title>Student Registration</title>
    <link rel="stylesheet" href="/style.css" />
  </head>

  <body>
    <div class="container">
```

```
<h1>Student Registration Form</h1>
<form action="/register" method="POST">
  <label>Student Name</label>
  <input type="text" name="name" required />

  <label>Register Number</label>
  <input type="text" name="regno" required />
  <label>Email</label>
  <input type="email" name="email" required />
  <label>Mobile Number</label>
  <input
    type="text"
    name="mobile"
    pattern="[0-9]{10}"
    maxlength="10"
    required/>
  <label>Gender</label>
  <div class="radio">
    <input type="radio" name="gender" value="Male" required />
    Male
    <input type="radio" name="gender" value="Female" />
    Female

    <input type="radio" name="gender" value="Other" />
    Other
  </div>
  <label>Date of Birth</label>
  <input type="date" name="dob" required />

  <label>Department</label>

  <select name="department" required>
    <option value="">Select Department</option>
    <option value="Computer Applications">Computer Applications</option>
    <option value="Computer Science">Computer Science</option>
    <option value="Information Technology">Information Technology</option>
```

```
  <option value="Electronics">Electronics</option>
</select>
<label>Course</label>

<select name="course" required>
  <option value="">Select Course</option>
  <option value="MCA">MCA</option>
  <option value="M.Sc Computer Science">M.Sc Computer Science</option>
  <option value="BCA">BCA</option>
  <option value="B.Sc Computer Science">B.Sc Computer Science</option>
</select>
<label>Year</label>
<select name="year" required>
  <option value="">Select Year</option>
  <option value="1st Year">1st Year</option>
  <option value="2nd Year">2nd Year</option>
  <option value="3rd Year">3rd Year</option>
</select>
<label>Address</label>
<textarea name="address" rows="4" required></textarea>

<label>Skills</label>
<div class="skills">
  <label>
    <input type="checkbox" name="skills" value="Python" />
    Python
  </label>

  <label>
    <input type="checkbox" name="skills" value="Java" />
    Java
  </label>

  <label>
    <input type="checkbox" name="skills" value="JavaScript" />
    JavaScript
  </label>

  <label>
```

```
          <input type="checkbox" name="skills" value="HTML/CSS" />
          HTML/CSS
        </label>

        <label>
          <input type="checkbox" name="skills" value="Node.js" />
          Node.js
        </label>
      </div>

      <button type="submit">Register Student</button>
    </form>

    <a class="logout" href="/logout">Logout</a>
  </div>
 </body>
</html>
```

Student.html

```
<!doctype html>
<html>
 <head>
  <title>Student Details</title>
  <link rel="stylesheet" href="/style.css" />
 </head>

 <body>
  <div class="container">
   <h1>Student Registration Successful</h1>

   <div id="studentDetails">Loading student details...</div>

   <br />
   <a class="button" href="/register"> Register Another Student </a>

   <a class="logout" href="/logout"> Logout </a>
  </div>

  <script>
   fetch("/student-data")
    .then((response) => response.json())
    .then((student) => {
     document.getElementById("studentDetails").innerHTML = `
```

```
<table>

   <tr>
      <th>Student Name</th>
      <td>${student.name}</td>
   </tr>
   <tr>
      <th>Register Number</th>
      <td>${student.regno}</td>
   </tr>

   <tr>
      <th>Email</th>
      <td>${student.email}</td>
   </tr>

   <tr>
      <th>Mobile Number</th>
      <td>${student.mobile}</td>
   </tr>
   <tr>
      <th>Gender</th>
      <td>${student.gender}</td>
   </tr>
   <tr>
      <th>Date of Birth</th>
      <td>${student.dob}</td>
   </tr>

   <tr>
      <th>Department</th>
      <td>${student.department}</td>
   </tr>

   <tr>
      <th>Course</th>
      <td>${student.course}</td>
   </tr>

   <tr>
```

```
            <th>Year</th>
            <td>${student.year}</td>
          </tr>

          <tr>
            <th>Address</th>
            <td>${student.address}</td>
          </tr>
          <tr>
            <th>Skills</th>
            <td>${student.skills}</td>
          </tr>
        </table>
      });
  </script>
 </body>
</html>
```

Style.css

```
body {
 font-family: Arial, sans-serif;
 background: #f2f2f2;
 margin: 0;
 padding: 30px;
}
.container {
 width: 650px;
 max-width: 90%;
 margin: auto;
 background: white;
 padding: 30px;
 border-radius: 10px;
 box-shadow: 0 0 10px #aaa;
}

h1 {
 text-align: center;
 margin-bottom: 25px;
}

h2 {
```

```
  text-align: center;
}
label {
  display: block;
  margin-top: 12px;
  font-weight: bold;
}
input,
select,
textarea {
  width: 100%;
  padding: 10px;
  margin-top: 5px;
  box-sizing: border-box;
  border: 1px solid #aaa;
  border-radius: 5px;
}

.radio {
  margin-top: 8px;
}

.radio input {
  width: auto;
  margin-right: 5px;
}
.skills label {
  font-weight: normal;
}

.skills input {
  width: auto;
}

button,
.button {
  display: block;
  width: 100%;
  padding: 12px;
  margin-top: 20px;
  background: #333;
```

```
  color: white;
  border: none;
  border-radius: 5px;
  text-align: center;
  text-decoration: none;
  cursor: pointer;
}
button:hover,
.button:hover {
  background: #555;
}

.logout {
  display: block;
  text-align:  center;
  margin-top: 20px;
  color: red;
}

.info {
  text-align: center;
  margin-top: 20px;
}

table {
  width: 100%;
  border-collapse: collapse;
}

th,
td {
  border: 1px solid #aaa;
  padding: 12px;
  text-align: left;
}

th {
  width: 35%;
}
```

App.js

```
const express = require("express");
const session = require("express-session");
const path = require("path");

const app = express();
const PORT = 3000;
// Middleware
app.use(express.urlencoded({ extended: true }));
app.use(express.static(path.join(__dirname, "public")));
app.use(
  session({
    secret: "student-registration-secret",
    resave: false,
    saveUninitialized: false,
  }),
);

// Login page
app.get("/", (req, res) => {
  res.sendFile(path.join(__dirname, "views", "login.html"));
});

// Login
app.post("/login", (req, res) => {
  const { username, password } = req.body;

  if (username === "admin" && password === "1234") {
    req.session.loggedIn = true;
    res.redirect("/register");
  } else {
    res.send(`
          <h2>Invalid username or password</h2>
          <a href="/">Back to Login</a>
        `);
  }
});
// Registration page
app.get("/register", (req, res) => {
if (!req.session.loggedIn) {
```

```
    return res.redirect("/");
  }
  res.sendFile(path.join(__dirname, "views", "register.html"));
});

// Handle registration
app.post("/register", (req, res) => {
if (!req.session.loggedIn) {
    return res.redirect("/");
  }
  const {
    name,
    regno,
    email,
    mobile,
    gender,
    dob,
    department,
    course,
    year,
    address,
    skills,
  } = req.body;

  // Validation
  if (
    !name ||
    !regno ||
    !email ||
    !mobile ||
    !gender ||
    !dob ||
    !department ||
    !course ||
    !year ||
    !address
  ) {
    return res.send(`
          <h2>All required fields must be filled.</h2>
          <a href="/register">Go Back</a>
        `);
  }
```

```
  // Email validation
  const emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;

  if (!emailPattern.test(email)) {
    return res.send(`
          <h2>Invalid email address.</h2>
          <a href="/register">Go Back</a>
        `);
  }

  // Mobile validation
  const mobilePattern = /^[0-9]{10}$/;
  if (!mobilePattern.test(mobile)) {
    return res.send(`
          <h2>Mobile number must contain exactly 10 digits.</h2>
          <a href="/register">Go Back</a>
        `);
  }
  // Store student details in session
  req.session.student = {
    name,
    regno,
    email,
    mobile,
    gender,
    dob,
    department,
    course,
    year,
    address,
    skills: skills || "None",
  };
  res.redirect("/student");
});

// Display student details
app.get("/student", (req, res) => {
if (!req.session.loggedIn) {
    return res.redirect("/");
  }
```

```
  if (!req.session.student) {
    return res.redirect("/register");
  }

  const student = req.session.student;

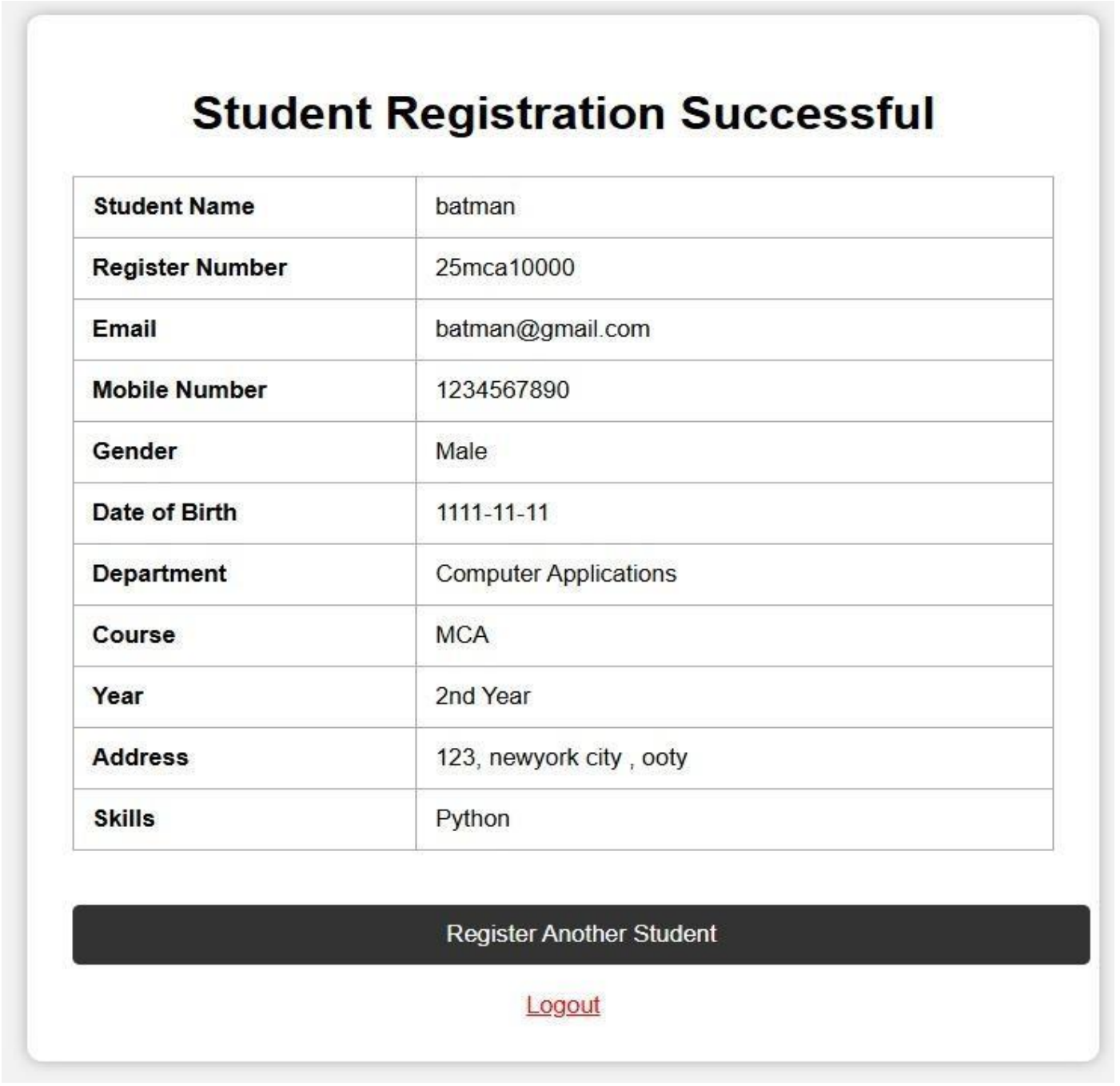
  res.sendFile(path.join(__dirname, "views", "student.html"));
});

// Student data API used by student.html
app.get("/student-data", (req, res) => {
if (!req.session.loggedIn) {
    return res.status(401).json({
      error: "Unauthorized",
    });
  }

  res.json(req.session.student);
});
// Logout
app.get("/logout", (req, res) => {
  req.session.destroy(() => {
  res.redirect("/");
  });
});

app.listen(PORT, () => {
  console.log(`Server running at http://localhost:${PORT}`);
});
```

Output



Student Registration Successful

Student Name	batman
Register Number	25mca10000
Email	batman@gmail.com
Mobile Number	1234567890
Gender	Male
Date of Birth	1111-11-11
Department	Computer Applications
Course	MCA
Year	2nd Year
Address	123, newyork city , ooty
Skills	Python

Register Another Student

Logout

7.Employee Information Portal using Node.js, Express.js and MongoDB

Source code

Employee.js

```
const mongoose = require("mongoose");
const employeeSchema = new mongoose.Schema({
  empId: { type: String, required: true, unique: true },
  name: { type: String, required: true },
  email: { type: String, required: true },
  phone: { type: String, required: true },
  department: { type: String, required: true },
  designation: { type: String, required: true },
  salary: { type: Number, required: true },
  joiningDate: { type: Date, required: true },
  address: { type: String },
  skills: { type: [String] }, // Array of strings
});

module.exports = mongoose.model("Employee", employeeSchema);
```

admin.js

```
const mongoose = require("mongoose");
const adminSchema = new mongoose.Schema({
  username: { type: String, required: true, unique: true },
  password: { type: String, required: true },
});

module.exports = mongoose.model("Admin", adminSchema);
```

server.js

```
require("dotenv").config();
const express = require("express");
const mongoose = require("mongoose");
const session = require("express-session");
const bcrypt = require("bcryptjs");
const Admin = require("./models/Admin");
const Employee = require("./models/Employee");

const app = express();

// Middleware
app.set("view engine", "ejs");
```

```
app.use(express.urlencoded({ extended: true }));
app.use(
  session({
    secret: "my-super-secret-key", // Replaced with a hardcoded string
    resave: false,
    saveUninitialized: false,
  }),
);

// MongoDB Connection
mongoose
  .connect(process.env.MONGO_URI, {
    useNewUrlParser: true,
    useUnifiedTopology: true,
  })
  .then(async () => {
    console.log("Connected to MongoDB");
    const adminExists = await Admin.findOne({ username: "admin" });
    if (!adminExists) {
      const hashedPassword = await bcrypt.hash("password123", 10);
      await Admin.create({ username: "admin", password: hashedPassword });
      console.log(
        "Default admin created (Username: admin, Password: password123)",
      );
    }
  })
  .catch((err) => console.log(err));

// Authentication Middleware
const requireAuth = (req, res, next) => {
  if (req.session.adminId) {
    next();
  } else {
    res.redirect("/login");
  }
};

// --- ROUTES ---

app.get("/login", (req, res) => res.render("login", { error: null }));

app.post("/login", async (req, res) => {
```

```
  const { username, password } = req.body;
  const admin = await Admin.findOne({ username });
  if (admin && (await bcrypt.compare(password, admin.password))) {
    req.session.adminId = admin._id;
    res.redirect("/");
  } else {
    res.render("login", { error: "Invalid username or password" });
  }
});

app.get("/logout", (req, res) => {
  req.session.destroy();
  res.redirect("/login");
});

// Dashboard
app.get("/", requireAuth, async (req, res) => {
  const { search, department } = req.query;
  let query = {};
  if (search) query.name = new RegExp(search, "i");
  if (department) query.department = department;

  const employees = await Employee.find(query).sort({ joiningDate: -1 });
  const departments = await Employee.distinct("department");
  res.render("dashboard", {
    employees,
    departments,
    searchQuery: search,
    deptQuery: department,
  });
});
// Add Employee
app.get("/add", requireAuth, (req, res) => res.render("add"));

app.post("/add", requireAuth, async (req, res) => {
  try {
    const newEmployee = new Employee({
      ...req.body,
      skills: req.body.skills.split(",").map((skill) => skill.trim()),
    });
    await newEmployee.save();
```

```
    res.redirect("/");
  } catch (err) {
    res.send("Error saving employee. Ensure Emp ID is unique.");
  }
});

// View Single Employee (NEW)
app.get("/view/:id", requireAuth, async (req, res) => {
  const employee = await Employee.findById(req.params.id);
  if (!employee) return res.status(404).send("Employee not found");
  res.render("view", { employee });
});

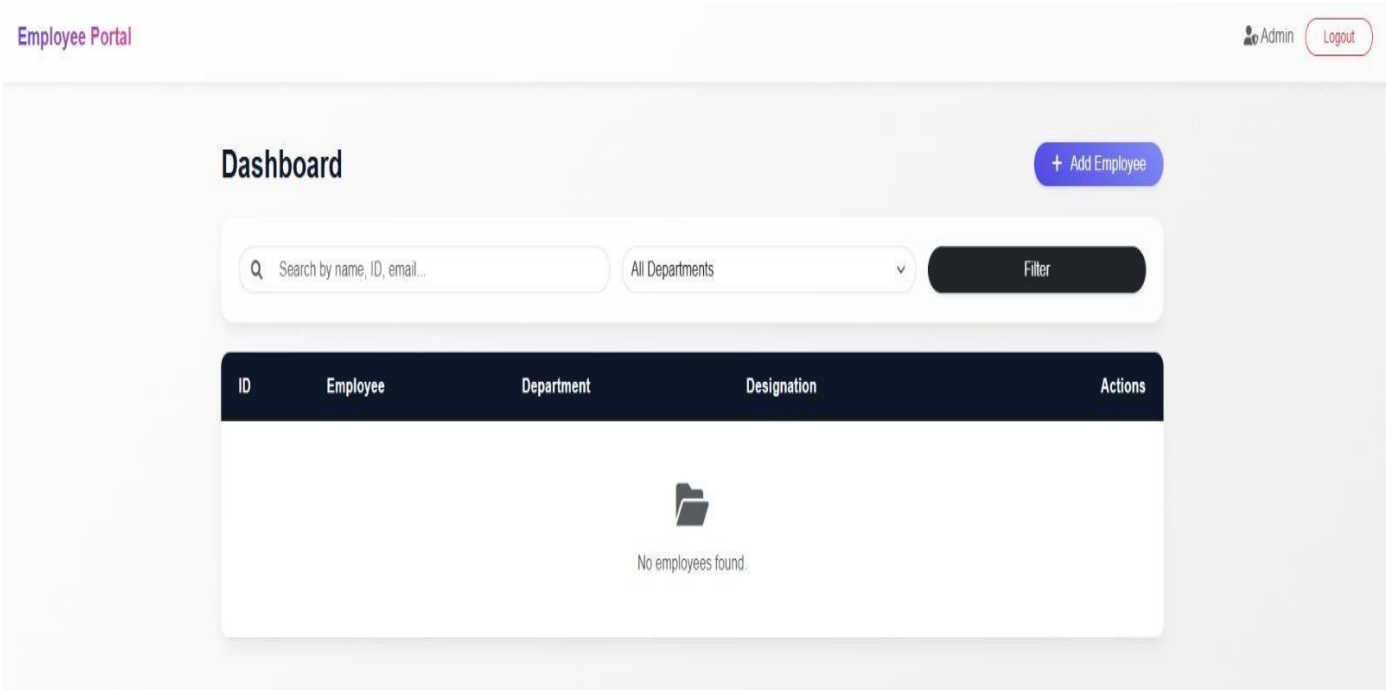
// Edit Employee
app.get("/edit/:id", requireAuth, async (req, res) => {
  const employee = await Employee.findById(req.params.id);
  res.render("edit", { employee });
});

app.post("/edit/:id", requireAuth, async (req, res) => {
  await Employee.findByIdAndUpdate(req.params.id, {
    ...req.body,
    skills: req.body.skills.split(",").map((skill) => skill.trim()),
  });
  res.redirect("/");
});

// Delete Employee
app.post("/delete/:id", requireAuth, async (req, res) => {
  await Employee.findByIdAndDelete(req.params.id);
  res.redirect("/");
});

const PORT = process.env.PORT || 3000;
app.listen(PORT, () =>
  console.log(`Server running on http://localhost:${PORT}`),
);
```

Output


Employee Portal
Admin
Logout
Dashboard
+ Add Employee
Search by name, ID, email...
All Departments
Filter
ID
Employee
Department
Designation
Actions
No employees found.

8. Git Repository: Basic Version Control Operations

Git Commands

git init

git add .

git commit -m "Initial commit"

git branch -M main

git remote add origin <GitHub-URL>

git push -u origin main

Output

```
$ git init
Reinitialized existing Git repository in C:/Users/rathi/OneDrive/Desktop/Opensourse Lab/Program1/.git/

rathi@hp MINGW64 ~/OneDrive/Desktop/Opensourse Lab/Program1 (master)
$ git add .

rathi@hp MINGW64 ~/OneDrive/Desktop/Opensourse Lab/Program1 (master)
$ git commit -m "first commit"
[master (root-commit) 49f201d] first commit
 7 files changed, 45 insertions(+)
 create mode 100644 app.js
 create mode 100644 assets/img1.jpg
 create mode 100644 assets/img2.jpg
 create mode 100644 assets/img3.jpg
 create mode 100644 assets/img4.avif
 create mode 100644 index.html
 create mode 100644 style.css

rathi@hp MINGW64 ~/OneDrive/Desktop/Opensourse Lab/Program1 (master)
$ ^C

rathi@hp MINGW64 ~/OneDrive/Desktop/Opensourse Lab/Program1 (master)
$ git branch -M main
```

```
rathi@hp MINGW64 ~/OneDrive/Desktop/Opensourse Lab/Program1 (main)
$

rathi@hp MINGW64 ~/OneDrive/Desktop/Opensourse Lab/Program1 (main)
$ git remote add origin https://github.com/rathidevikarthikeyan/Slide-carousel_01.git
git branch -M main
git push -u origin main
error: remote origin already exists.
info: please complete authentication in your browser...
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 12 threads
Compressing objects: 100% (10/10), done.
Writing objects:  40% (4/10), 1.49 MiB | 68.00 KiB/s
```

